

Maximum Marks: 100

EE228 – Electrical Network Analysis (EE-1, 2, 3)

FINAL EXAM (SPRING 2016)

Total Time: 180 min

Instructors: Dr. Anjum Ali & Mr. Mohsin Yousuf

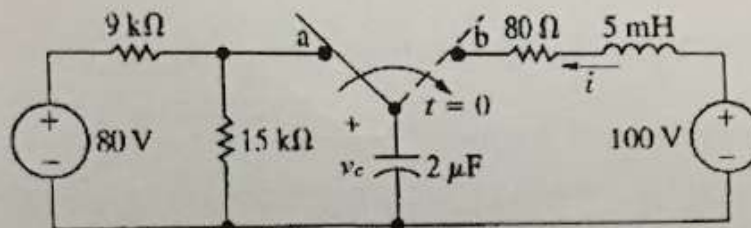
Dated: May 9, 2016

Attempt all questions with proper labeling of parts.
Total number of pages: 3

$$\Delta = \frac{1}{2RC}$$

Question # 1 [4+4+4+4+4 = 20 marks]

The switch in the circuit shown below has been in position 'a' for a long time. At time $t = 0$, it moves to position 'b'.



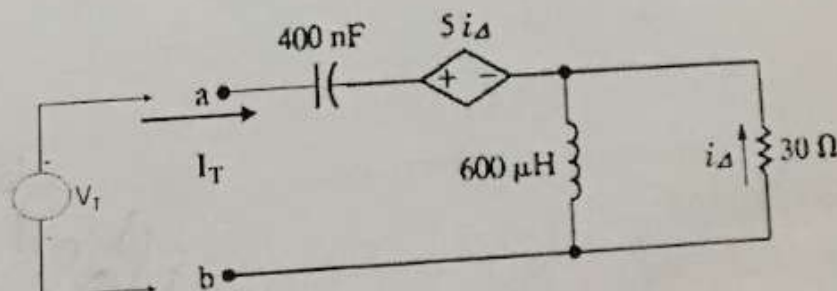
$$\begin{aligned} i(t) &= A_1 e^{s_1 t} + A_2 e^{s_2 t} \text{ (overdamped),} \\ i(t) &= B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t \text{ (underdamped),} \\ i(t) &= D_1 t e^{-\alpha t} + D_2 e^{-\alpha t} \text{ (critically damped).} \end{aligned}$$

$$\begin{aligned} v_C &= V_f + A'_1 e^{s_1 t} + A'_2 e^{s_2 t} \text{ (overdamped),} \\ v_C &= V_f + B'_1 e^{-\alpha t} \cos \omega_d t + B'_2 e^{-\alpha t} \sin \omega_d t \text{ (underdamped),} \\ v_C &= V_f + D'_1 t e^{-\alpha t} + D'_2 e^{-\alpha t} \text{ (critically damped).} \end{aligned}$$

- Find $i(0+)$, $v_C(0+)$, $\frac{di(0+)}{dt}$ and $\frac{dv_C(0+)}{dt}$.
- Write the second order differential equation for $i(t)$ in time domain for $t \geq 0$ and find the characteristic equation as well.
- Give the roots of the characteristic equation and identify the type of response from the roots.
- Give the complete expression for $i(t)$ and $v_C(t)$ for $t \geq 0$.
- Identify the transient and steady-state part of $i(t)$ and $v_C(t)$.

Question # 2 (a) [10 marks]

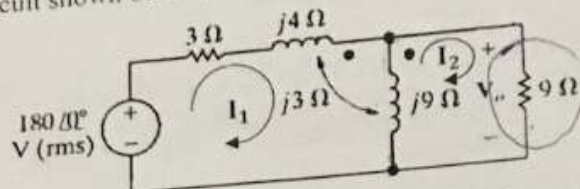
Find Z_{ab} , using test source method in the circuit shown below. The circuit is operating at a frequency of $\omega = 100 \text{ krad/s}$. Hint: Assume sinusoidal voltages and currents, and convert the circuit to phasor domain first.



$$-\alpha - \frac{1}{2RC}$$

Question # 2 (b) [10 marks]

For the frequency-domain circuit shown below, find the following:



- Using mesh analysis, write the two mesh equations for I_1 and I_2 .
- Find V_o (rms).
- Find the average power dissipated in the 9Ω resistor.

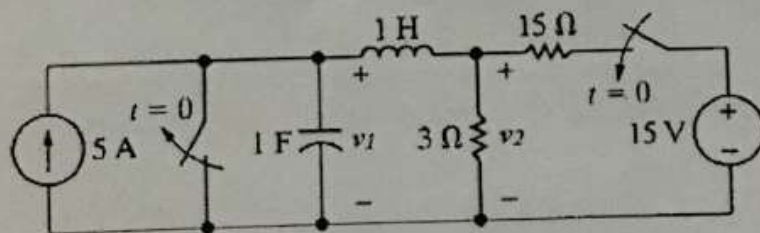
Question # 3 [8+4+8 = 20 marks]

A Y- Δ connected balanced three phase system is characterized as follows:

- Internal impedance of the generator = $0 + j0 \Omega/\phi$;
 - Source voltage in the c-phase is $20\angle -90^\circ$ V;
 - Source phase sequence is abc;
 - Line impedance is $1 + j3 \Omega/\phi$;
 - Load impedance is $117 - j99 \Omega/\phi$.
- Draw the single phase equivalent for the a-phase.
 - Calculate the a-phase line current.
 - Calculate the a-phase line voltage at the load.

Question # 4 [10+6+2+2 = 20 marks]

The dc current and voltage sources are applied simultaneously to the circuit shown at $t = 0$. No energy is stored in the circuit at the instant of application.



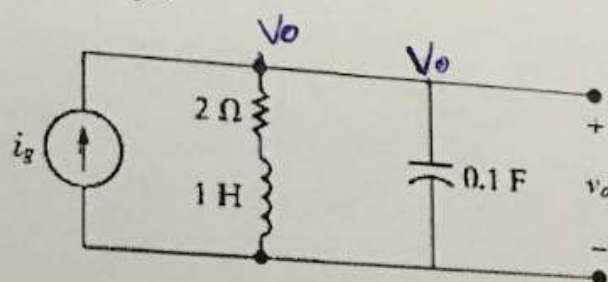
- Using NODE analysis, find the s-domain expressions for V_1 and V_2 .
- For $t > 0$, find the time-domain expressions for v_1 and v_2 .
- Calculate $v_1(0+)$ and $v_2(0+)$.
- Compute the steady-state values of v_1 and v_2 .

Handwritten notes: $s^2 \times \left(\frac{1}{RC}\right)s \times \frac{1}{\omega^2 0}$

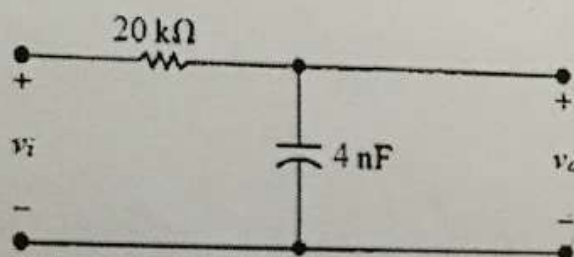
Question # 5 [10+10 = 20 marks]

- (a) Find the transfer function, $H(s)$, and the unit-impulse response, $h(t)$, for the following circuit.

Hint: $h(t) = \mathcal{L}^{-1}[H(s)]$ and $H(s) = \frac{V_o(s)}{I_g(s)}$



- (b) For the filter circuit shown below, answer the following:



- (i) Is it a high-pass or a low-pass filter?
- (ii) Find the cut-off frequency (in hertz).
- (iii) Find $H(s)$ and calculate $H(j\omega)$ at ω_c , $0.2\omega_c$, and $8\omega_c$.
- (iv) If $v_i = 0.48 \cos \omega t$ V, write the steady-state expression for v_o when $\omega = 8\omega_c$.

Hint: $v_{o_{ss}}(t) = A|H(j\omega)| \cos[\omega t + \theta(\omega)]$